

Interacting Multipoles of Rank 2 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}\cdot$$

$$\begin{aligned} &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\gamma,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\epsilon)\cdot\delta(\gamma,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\beta)\cdot\delta(\epsilon,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\epsilon)\cdot\delta(\beta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \end{aligned}$$

$$(1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}.$$

[illegible]

[illegible]

[illegible]

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}.$$

[illegible]

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1890 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -3780 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} \\ -3780 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} \\ +210 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +210 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} \\ -15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \end{bmatrix} \quad (3c)$$

Simplify Quadrupoles:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta} \Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1890 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -3780 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\ -3780 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ +210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} \\ -15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1890 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ +210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} \\ -15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{bmatrix} \quad (4b)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ +210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} \\ -15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{bmatrix} \quad (4c)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta} \Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{bmatrix} \quad (5)$$

Reduce Indices:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} \\ -1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} \\ -1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \quad (6a)$$

$$\xRightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -3150 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} \\ +315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \quad (6b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = \begin{bmatrix} -10 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} \\ +1 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +4 \cdot R_z^{-7} \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -\frac{4}{7} \cdot R_z^{-7} \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \quad (7)$$